

**SIMATS ENGINEERING
ASSESSMENT TOOL 4**

COURSE NAME: Artificial Intelligence

COURSE CODE: CSA17

COURSE OUTCOME COVERED

CO3: *Analyze knowledge representation and reasoning using propositional logic, FOL, inference, resolution, chaining methods, knowledge representation to perform logical reasoning for drawing valid conclusions and decision making. (BL4)*

Assessment Tool Weightage: 20%

QUESTIONS: 5

**Time Duration: 1 Hour
Total Marks:30**

Annexure A

Resolution Algorithm Implementation

Code of Conduct:

I Arava Raja (192365033) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.

I understand that any violation of academic integrity rules will result in disciplinary action.

Arava Raja

Signature of the student:

Resolution Algorithm Implementation

Question 1 – Rain and Wet Ground

Knowledge Base

1. If it rains, then the ground becomes wet.
2. It is raining.

Goal

Prove that **the ground is wet** using the Resolution Algorithm.

Tasks

1. Represent the statements in propositional logic.
2. Convert all statements into CNF.

Negate the goal.

1. Apply the Resolution Algorithm step by step.
2. Show the Empty Clause ($\square$) if the goal is proved.

Question 2 – Student Assignment Submission

Knowledge Base

1. If a student submits the assignment, then the student receives internal marks
2. Rahul submitted the assignment.

Goal

Prove that **Rahul receives internal marks** using the Resolution Algorithm.

Tasks

1. Write the propositional logic expressions.
2. Convert the premises into CNF.

Negate the goal.

1. Apply the Resolution Algorithm.
2. Conclude whether the goal is proved.

Question 3 – Library Membership

Knowledge Base

1. If a person is a library member, then the person can borrow books.
2. Priya is a library member.

Goal

1. Prove that **Priya can borrow books** using the Resolution Algorithm.

Tasks

1. Represent the knowledge base using propositional logic.
2. Convert each statement into CNF.

Negate the goal.

1. Perform resolution until no new clauses can be generated.
2. State the final conclusion.

Question 4 – Placement Eligibility

Knowledge Base

1. If a student clears the aptitude test, then the student is eligible for placement.
2. Arun cleared the aptitude test.

Goal

1. Prove that **Arun is eligible for placement** using the Resolution Algorithm.

Tasks

1. Convert the statements into propositional logic.
2. Write the CNF clauses.
3. Negate the goal.
4. Apply the Resolution Algorithm step by step.
5. Draw the resolution tree and conclude.

Question 5 – Access Control System

Knowledge Base

1. If a user enters the correct password, then the user is authenticated.
2. If a user is authenticated, then the user is granted access.
3. The user entered the correct password.

Goal

Prove that **the user is granted access** using the Resolution Algorithm.

Tasks

1. Represent the knowledge base in propositional logic.
2. Convert all implications into CNF.
3. Negate the goal.
4. Apply the Resolution Algorithm to derive the Empty Clause ($\square$).
5. Explain each resolution step and write the final conclusion.

1.

RUBRICS FOR CALCULATION

Numerical Problems					
Criteria	Weightage	Level 4 (Exemplary)	Level 3 (Proficient)	Level 2 (Developing)	Level 1 (Beginning)
Problem Understanding	20%	Clearly interprets problem, identifies all parameters and requirements accurately	Understands problem with minor gaps	Partial understanding; misses key elements	Misinterprets or unable to understand problem
Method Selection	20%	Selects most appropriate method/formula with strong justification	Selects correct method with minor errors	Inappropriate or partially correct method	Unable to select method
Solution Process	25%	Logical, systematic steps with correct approach throughout	Mostly correct steps with minor mistakes	Disorganized steps; multiple errors	No clear process
Accuracy of Calculation	20%	All calculations accurate; correct final answer	Minor calculation errors	Frequent errors; incorrect final answer	No valid calculations
Interpretation of Results	15%	Clearly interprets results with units and engineering relevance	Basic interpretation with minor omissions	Limited or unclear interpretation	No interpretation

Question 1 – Rain and Wet Ground

Given Knowledge Base

1. If it rains, then the ground becomes wet.
2. It is raining.

Goal

Prove that the ground is wet using the Resolution Algorithm.

1. Propositional Logic

Let:

- **R** = It is raining.
- **W** = The ground is wet.

Statement 1:

$R \rightarrow W$

Statement 2:

R

Goal:

W

2. Convert to CNF

Using:

$P \rightarrow Q \equiv \neg P \vee Q$

Therefore,

$R \rightarrow W$

becomes:

$\neg R \vee W$

The clauses are:

- **C1: $\neg R \vee W$**
- **C2: R**

3. Negate the Goal

Goal = **W**

Negated goal:

$\neg W$

So the clauses are:

- **C1: $\neg R \vee W$**
- **C2: R**
- **C3: $\neg W$**

4. Resolution Steps

Resolve C1 and C2:

C1: $\neg R \vee W$

C2: R

Resolving **$\neg R$** and **R**:

C4: W

Now resolve C4 with the negated goal C3:

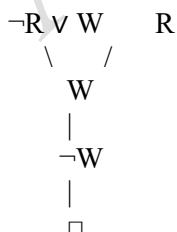
C4: W

C3: $\neg W$

Resolving **W** and **$\neg W$** gives:

□

Resolution Tree



Conclusion

Since the resolution process produces the **Empty Clause (□)**, the negation of the goal leads to a contradiction.
Therefore:

The ground is wet. ✓

Question 2 – Student Assignment Submission

The question states that if a student submits the assignment, the student receives internal marks, and Rahul submitted the assignment.

Given Knowledge Base

1. If a student submits the assignment, then the student receives internal marks.
2. Rahul submitted the assignment.

Goal

Prove that **Rahul receives internal marks.**

1. Propositional Logic

Let:

- **S** = Rahul submitted the assignment.
- **M** = Rahul receives internal marks.

Statement 1:

$S \rightarrow M$

Statement 2:

S

Goal:

M

2. CNF

Convert implication:

$S \rightarrow M \equiv \neg S \vee M$

Therefore:

- **C1: $\neg S \vee M$**
- **C2: S**

3. Negate the Goal

Goal:

M

Negated goal:

$\neg M$

Therefore:

- **C1: $\neg S \vee M$**
- **C2: S**
- **C3: $\neg M$**

4. Resolution

Resolve C1 and C2:

$\neg S \vee M$

S

Therefore:

C4: M

Now resolve C4 and C3:

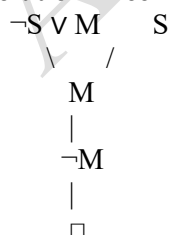
M

$\neg M$

Therefore:

□

Resolution Tree



Conclusion

The Empty Clause $\square$ is obtained. Hence the negation of the goal is contradictory.

Therefore:

Rahul receives internal marks. ✓

Question 3 – Library Membership

The question gives that a library member can borrow books and that Priya is a library member.

Given Knowledge Base

1. If a person is a library member, then the person can borrow books.
2. Priya is a library member.

Goal

Prove that **Priya can borrow books.**

1. Propositional Logic

Let:

- **L** = Priya is a library member.
- **B** = Priya can borrow books.

Statement 1:

$L \rightarrow B$

Statement 2:

L

Goal:

B

2. CNF

Convert:

$L \rightarrow B$

Therefore:

$\neg L \vee B$

Clauses:

- **C1: $\neg L \vee B$**
- **C2: L**

3. Negate the Goal

Goal:

B

Negated goal:

$\neg B$

Thus:

- **C1: $\neg L \vee B$**
- **C2: L**
- **C3: $\neg B$**

4. Resolution

Resolve C1 and C2:

$\neg L \vee B$

L

Therefore:

C4: B

Resolve C4 and C3:

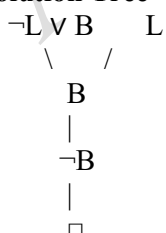
B

$\neg B$

Therefore:

□

Resolution Tree



Final Conclusion

Since the Empty Clause □ is derived, the negation of the goal is false.

Therefore:

Priya can borrow books. ✓

Question 4 – Placement Eligibility

The question states that clearing the aptitude test makes a student eligible for placement, and Arun has cleared the aptitude test.

Given Knowledge Base

1. If a student clears the aptitude test, then the student is eligible for placement.
2. Arun cleared the aptitude test.

Goal

Prove that **Arun is eligible for placement.**

1. Propositional Logic

Let:

- **A** = Arun cleared the aptitude test.
- **E** = Arun is eligible for placement.

Statement 1:

$$A \rightarrow E$$

Statement 2:

$$A$$

Goal:

$$E$$

2. CNF Clauses

Convert the implication:

$$A \rightarrow E$$

Using:

$$P \rightarrow Q \equiv \neg P \vee Q$$

Therefore:

$$\neg A \vee E$$

Clauses:

- **C1:** $\neg A \vee E$
- **C2:** **A**

3. Negate the Goal

Goal:

$$E$$

Negated goal:

$$\neg E$$

Therefore:

- **C1:** $\neg A \vee E$
- **C2:** **A**
- **C3:** $\neg E$

4. Resolution Steps

Resolve C1 and C2:

$$\mathbf{C1: \neg A \vee E}$$

$$\mathbf{C2: A}$$

The complementary literals $\neg A$ and **A** cancel.

Therefore:

$$\mathbf{C4: E}$$

Now resolve C4 with C3:

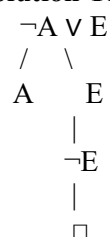
$$\mathbf{C4: E}$$

$$\mathbf{C3: \neg E}$$

Resolving **E** and $\neg E$:

□

5. Resolution Tree



Conclusion

The Empty Clause $\square$ has been derived. Therefore, the negation of the goal is contradictory.

Hence:

Arun is eligible for placement. ✓

Question 5 – Access Control System

The question gives two implications: correct password $\rightarrow$ authenticated, and authenticated $\rightarrow$ access, along with the fact that the user entered the correct password.

Given Knowledge Base

1. If a user enters the correct password, then the user is authenticated.
2. If a user is authenticated, then the user is granted access.
3. The user entered the correct password.

Goal

Prove that **the user is granted access**.

1. Propositional Logic

Let:

- P = User entered the correct password.
- A = User is authenticated.
- G = User is granted access.

The statements are:

$P \rightarrow A$

$A \rightarrow G$

P

Goal:

G

2. Convert to CNF

First implication:

$P \rightarrow A$

becomes:

$\neg P \vee A$

Second implication:

$A \rightarrow G$

becomes:

$\neg A \vee G$

Third statement:

P

Therefore, the CNF clauses are:

- $C1: \neg P \vee A$
- $C2: \neg A \vee G$
- $C3: P$

3. Negate the Goal

Goal:

G

Negated goal:

$\neg G$

Therefore, our complete clause set is:

- $C1: \neg P \vee A$
- $C2: \neg A \vee G$
- $C3: P$
- $C4: \neg G$

4. Resolution Step 1

Resolve $C1$ and $C3$:

$C1: \neg P \vee A$

$C3: P$

The complementary literals $\neg P$ and P cancel.

Therefore:

C5: A

5. Resolution Step 2

Resolve C2 and C5:

C2: $\neg A \vee G$

C5: A

The complementary literals $\neg A$ and A cancel.

Therefore:

C6: G

6. Resolution Step 3

Resolve C6 with the negated goal C4:

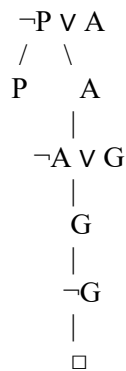
C6: G

C4: $\neg G$

Therefore:

□

Resolution Tree



Final Conclusion

The resolution procedure produces the **Empty Clause** ($\square$).

Therefore, the negation of the goal $\neg G$ is contradictory.

Hence:

The user is granted access.